

choptuik_ac_bc · theorems12 · github.com/wild8highlander/choptuik_ac_bc

The Figure-Cycle Predictor

Theorem: the pyramid–cone–bowl cycle carries global components of S

36 compensated states inside S · 12/12 + 6/6 stations · return defect 0.064, period 6

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Abstract

This supplementary monograph records the figure-cycle predictor theorem of session v16 (machine `hexcycle_dae.py`, verdict `hexcycle_dae.json`, LEDGER §27): the Poincaré hexagonal figure cycle (pyramid $\rightarrow$ cone $\rightarrow$ bowl) embedded into the nonlinear DAE tower through the axioms E1–E3 carries global components of the static solution set S — 36 compensated states verified to lie in S ($F = r = 0$), answering the v15 caveat about components away from the critical point.

The compensated book lands 12/12 collapse stations and 6/6 blowup stations with the ruler returning to $R1h \approx 1$; the return defect per cycle is $\delta_{R1h} = 0.0642$ with a six-station periodicity $\max|R1h(k+6) - R1h(k)| = 0.3085$; the station-0 residual is 5.7×10^{-14} (the cycle reproduces x^* exactly); the L1 landings with $|V| \neq 0$ confirm globally that the flow leaves S (exit rates $2.2 \times 10^{-14} \dots 7.9 \times 10^6$). Honest caveats: the figure profiles remain a visualization layer; the embeddings E1–E3 are axioms. Tables, corollaries and the reproduction protocol are included.

Keywords: hexcycle, figure cycle, global statics, compensated book, station landing, Newton multi-start, machine theorem.

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1. Introduction

This supplementary monograph records the figure-cycle predictor theorem — the machine result of session v16 (HEXCYCLE-DAE, machine `hexcycle_dae.py`, LEDGER §27). The v15 march theorem (monograph No. 6) established that the local march off the flat line is impossible and left one honest caveat: global components of the static set S away from all starts were not excluded. Session v20 (published as v16 of the ledger line) answered with a global construction: the hexagonal figure cycle of the programme, embedded through its machine anchors, lands compensated states that lie in S .

The theorem matters because it converts a caveat into a classification: the static world of the tower is not exhausted by the critical point and its line — the figure cycle supplies explicit global static components, while the kinetic layer confirms that the dynamics still leaves S for generic amplitudes. Both halves are needed: the statics is bigger than x^* , and the dynamics still does not live there.

2. Notation and Setting

The figure cycle is the hexagonal sequence of figure profiles (pyramid $\rightarrow$ cone $\rightarrow$ bowl) whose machine anchors are the clock ratios $\tau_3 = (3/2)^2 = 2.25$ and $\tau_5 = 1/(4/3)^2 = 0.5625$, with $\kappa_{\text{cyc}} = \ln(\tau_3/\tau_5^2) = 1.9616585060$ and $\Delta_{\text{cyc}} = 6\ln(16/9) = 3.4521848694$. The embedding axioms: (E1) $T0h_k = T0h^* \cdot \exp(\mp r_k)$ with r_k the accumulated steps; station 0 is the critical scale; (E2) pure reading $R1h = 1$, the ruler compensation is tested separately ($T1c/T1d$); (E3) amplitudes are the exact v15 chain forms ($D0h = 0$) with homothetic kinematics $vT = 1/(2T0h)$ — the prolongation holds by construction.

The setting: land the compensated book on the cycle stations (Newton multi-start per station), verify the landed states against the full nonlinear system (F, r, V) , measure the ruler-return defect and its periodicity, and test the L1 landings with nonzero velocity for the exit verdict.

Table 14.1. The cycle anchors (machine axioms layer)

Anchor	Value	Meaning
τ_3	$2.25 = (3/2)^2$	the pyramid clock ratio
τ_5	$0.5625 = (3/4)^2$	the bowl clock ratio
κ_{cyc}	1.9616585060	$\ln(\tau_3/\tau_5^2)$ — the cycle book of κ
Δ_{cyc}	3.4521848694	$6\ln(16/9)$ — the DSS normalization

3. Statement of the Theorem

Theorem 14 (the figure cycle as a global predictor). Under the embedding axioms E1–E3:

- (i) station 0 reproduces the critical point exactly: $T0h^* = 2.5980762114$, $\tau^* = 6.75$, residual 5.7×10^{-14} (station0_is_x_star = 3.6×10^{-15});
- (ii) the compensated book lands 12/12 collapse stations and 6/6 blowup stations (minimal correction, the degenerate branch rejected); the ruler returns along the cycle with the defect $\delta_{R1h} = 0.0642$ per cycle (collapse) and the six-station periodicity $\max |R1h(k+6) - R1h(k)| = 0.3085$;
- (iii) 36 compensated states lie in S : $F = r = 0$ with machine zero — global static components of the solution set are found (the answer to the v15 caveat); the dimension of the static set at pinned $T0h$ is $[0, 1]$;
- (iv) the static amplitude book is not exact away from the critical scale: the raw F_{max} over the cycle is 2.707×10^5 (station 0: 1.36×10^{-12}) — the compensation is not optional;
- (v) the dynamics leaves S globally: L1 landings with $|V| \neq 0$ give exit rates $2.185 \times 10^{-14} \dots 7.914 \times 10^6$, and the kinetic layer shows $r_{\text{max}} = 8.879$ — the closure is incompatible with the homothetic walking; the L1 global-shooting landings converge 14/24.

4. Sketch of the Machine Proof

The machine builds the station ladder from the anchors (r_k accumulated by the book steps), lands each station by a damped Newton on the compensated book with multistart

(the degenerate branch is rejected by the condition number), verifies each landed state against the full nonlinear F and the consistency r , and measures the ruler drift along the cycle. The kinetic layer marches the landed states with the homothetic kinematics and measures the exit rate of $|V| \neq 0$ amplitudes. All landings are deterministic (fixed seeds, $\text{BLAS} = 1$).

Table 14.2. The landing verdicts (verdict_lines)

Layer	Outcome
Compensated book (collapse)	12/12 stations; R1h returns: 1.012 $\rightarrow$ 0.687 $\rightarrow$ 0.720 $\rightarrow$ 0.845 $\rightarrow$ 0.918 $\rightarrow$ 0.936 $\rightarrow$ 0.974 $\rightarrow$ 0.995 $\rightarrow$ 0.998 $\rightarrow$ 0.999 $\rightarrow$ 0.999 $\rightarrow$ 1.0
Compensated book (blowup)	6/6 stations; R1h: 2.127 $\rightarrow$ 2.137 $\rightarrow$ 2.264 $\rightarrow$ 1.121 $\rightarrow$ 0.085 $\rightarrow$ 0.083
Global statics	36 compensated states in S ($F = r = 0$); dim at pinned $T0h = [0, 1]$
Ruler return	$\delta_{R1h} = 0.0642$ (1 cycle); periodicity $\max R1h(k+6) - R1h(k) = 0.3085$
Kinetic layer	$r_{\text{max}} = 8.879$; exit rates $2.2 \times 10^{-14} \dots 7.9 \times 10^6$; landings 14/24

5. Numerical Results

The collapse ladder shows the ruler drifting away from 1 at mid-cycle (down to 0.687 at station 1) and returning to 1.0 at the close — the six-station periodicity is the machine shadow of the hexagonal structure. The blowup ladder runs the inverse direction (2.127 $\rightarrow$ 0.083); the cycle is read backwards for the dispersing branch. The 36 compensated states in S are distributed along both ladders and certify that the static set is not exhausted by the flat line — the honest answer to the v15 caveat, with the caveat now replaced by an explicit construction.

The raw (uncompensated) static book fails off station 0 ($F_{\text{max}} = 2.707 \times 10^5$); this is the quantitative form of the statement that the amplitude book needs the compensation — the compensated book is the machine object, the raw profile is a visualization layer.

6. Corollaries

Corollary 14.1 (the static world is bigger than x^*). The flat line of monograph No. 6 is not the whole static set: the cycle supplies 36 explicit global components. Any future claim of static exhaustiveness must be phrased per class (isolated roots — monograph No. 9; non-isolated families — monographs No. 3 and No. 14).

Corollary 14.2 (the compensation is structural). The ruler-return defect 0.0642 with period 6 shows the compensation is a well-defined machine operation, not a fit: the same compensation lands both branches (collapse and blowup) with the degenerate branch cleanly rejected.

Corollary 14.3 (dynamics still leaves S). The exit rates of (v) confirm the $v15$ verdict globally: the compensated statics does not create a live carrier — the echo physics remains with the PDE machine (monographs No. 11–12).

7. Honest Caveats

The embedding axioms E1–E3 are axioms: the exponential step reading and the homothetic kinematics are the tested embedding, and alternative readings (non-exponential step, wandering R1h as part of the book) are covered only by the scan/landing layers. The landings are a local Newton analysis; global components of S away from all starts are not excluded — the theorem's claim is the existence of the 36 components, not exhaustiveness. The P4h pole crossing ($\tau = 5.625$) is a property of the solved branch; Puiseux-type branch jumps are not excluded, and the multistart barrier test is the guard. Finally, the figure profiles remain a visualization layer: the machine objects are the anchors, the stations and the landed states.

8. Reproducibility

Table 14.3. Reproduction protocol for Theorem 14

Step	Command	Result
1	<code>python3 hexcycle_dae.py</code>	<code>results/hexcycle_dae.json</code>

Step	Command	Result
		(landings, statics, kinetics)
2	python3 march_dae_nonlinear.py	results/ march_dae_nonlinear.json (the local verdict being globalized)
3	python3 -m pytest tests/ -q	58 passed (hexcycle tests included)

The machine runs in minutes (deterministic, single-threaded). The data correspond to commit 5b256cd; an autonomous copy lives in `einstein_direct/open9/machines/`.

References

- [1] Repository `wild8highlander/choptuik_ac_bc`: `einstein_direct/hexcycle_dae.py`;
`einstein_direct/results/hexcycle_dae.json` (session v16/v20, LEDGER §27).
- [2] Ibid.: `einstein_direct/dae_nonlinear_core.py`;
`einstein_direct/results/march_dae_nonlinear.json` (the v15 local verdict).
- [3] Monographs No. 3, No. 6, No. 9 of the theorems12 cycle.